NosAnalizamos

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R Markdown

LIBRERÍAS

```
library(readr)
```

```
## Warning: package 'readr' was built under R version 4.5.3
```

```
library(dplyr)
```

```
## Warning: package 'dplyr' was built under R version 4.5.3

##
## Adjuntando el paquete: 'dplyr'

## The following objects are masked from 'package:stats':
##
##     filter, lag

## The following objects are masked from 'package:base':
##
##     intersect, setdiff, setequal, union
```

```
library(lubridate)
```

```
## Warning: package 'lubridate' was built under R version 4.5.3

##
## Adjuntando el paquete: 'lubridate'

## The following objects are masked from 'package:base':
##
##     date, intersect, setdiff, union
```

```
library(stringr)
```

```
## Warning: package 'stringr' was built under R version 4.5.3
```

```
library(forcats)
```

```
## Warning: package 'forcats' was built under R version 4.5.3
```

```
library(tidyr)
```

```
## Warning: package 'tidyr' was built under R version 4.5.3
```

```
library(rticles)
```

```
## Warning: package 'rticles' was built under R version 4.5.3
```

```r
library(kableExtra)
```

```
## Warning: package 'kableExtra' was built under R version 4.5.3
```

```
##
## Adjuntando el paquete: 'kableExtra'
```

```
## The following object is masked from 'package:dplyr':
##
##     group_rows
```

```r
library(ggplot2)
```

```
## Warning: package 'ggplot2' was built under R version 4.5.3
```

```r
library(pracma)
```

```
## Warning: package 'pracma' was built under R version 4.5.3
```

```r
library(GGally)
```

```
## Warning: package 'GGally' was built under R version 4.5.3
```

CARGA DATOS FICHERO

```r
datos <- read_delim("data/NosAutoanalizamos2022 - Hoja1.tsv",
                    delim = "\t",
                    escape_double = FALSE, #""
                    trim_ws = TRUE, #Fuera espacios en blanco
                    skip = 25) %>%
  select(-1) %>%
  filter(!is.na(Id))
```

```
## New names:
## Rows: 64 Columns: 21
## -- Column specification
## -------------------------------------------------------- Delimiter: "\t" chr
## (15): ...1, FechaNac(DD-MM-YYY), Id, Sex, Wr.Hnd, NW.Hnd, Fold, Clap, Sm... dbl
## (5): Age, Pulse, Exer, Height, HSt num (1): Hwork
## i Use `spec()` to retrieve the full column specification for this data. i
## Specify the column types or set `show_col_types = FALSE` to quiet this message.
## * `` -> `...1`
```

Eliminar a usuario

```r
datos <- datos %>%
  filter(Id != "martsobm")
```

Modificar columnas de calificaciones

```r
datos <- datos %>%
  mutate(across(c(ALG, ANM, FP, DCS, MD), ~ {
    # 1. Convertir a texto para manipular
    x <- as.character(.x)
```

```r
    # 2. Sustituir coma por punto
    x <- str_replace(x, ",", ".")

    # 3. Sustituir "NC" por 3
    x[x == "NC"] <- "3"
    as.numeric(x) #Si no es numérico lo sustituye por NA
  }))
```

```
## Warning: There were 2 warnings in `mutate()`.
## The first warning was:
## i In argument: `across(...)`.
## Caused by warning:
## ! NAs introducidos por coerción
## i Run `dplyr::last_dplyr_warnings()` to see the 1 remaining warning.
```

```r
datos
```

```
## # A tibble: 62 x 20
##    `FechaNac(DD-MM-YYY)` Id      Age Sex   Wr.Hnd NW.Hnd Fold  Pulse Clap   Exer
##    <chr>                 <chr> <dbl> <chr> <chr>  <chr>  <chr> <dbl> <chr> <dbl>
##  1 18-11-2003            pede~    18 M     18     Derec~ Izqu~    50 Der ~     5
##  2 19-08-2003            salg~    18 M     19     Derec~ Dere~    73 Der ~     0
##  3 15-03-2003            garf~    18 F     18     Izqui~ Izqu~    71 Der ~     0
##  4 24-06-2003            silo~    18 F     18     Derec~ Dere~    66 Der ~     0
##  5 21-6-2003             jama~    18 M     16     Derec~ Dere~    60 Der ~     4
##  6 13-05-2003            maba~    18 M     20     Izqui~ Izqu~    96 Izq ~     6
##  7 6-9-2003              esaia    18 F     16     Derec~ Dere~    63 Der ~     3
##  8 09-01-2003            bepe~    19 F     14     Derec~ Izqu~    64 Der ~     3
##  9 14-05-2002            omeal    19 M     21     Derec~ Dere~    80 Der ~     7
## 10 30-08-2003            ensa~    18 M     22     Derec~ Dere~    80 Izq ~     3
## # i 52 more rows
## # i 10 more variables: Smoke <chr>, Height <dbl>, ALG <dbl>, ANM <dbl>,
## #   FP <dbl>, DCS <dbl>, MD <dbl>, HSt <dbl>, Hwork <dbl>, Comentarios <chr>
```

```r
# Guardar en Notas.Rdata como pide el enunciado
save(datos, file = "Notas.Rdata")
```

#Calcular tabla estadística

```r
# 1. Calcular los estadísticos en formato largo
tabla_resumen <- datos %>%
  select(ALG, ANM, FP, DCS, MD) %>%
  pivot_longer(cols = everything(), names_to = "Asignatura", values_to = "Nota") %>%
  group_by(Asignatura) %>%
  summarise(
    Minimo = min(Nota, na.rm = TRUE),
    Percentil25 = quantile(Nota, 0.25, na.rm = TRUE),
    Mediana = median(Nota, na.rm = TRUE),
    Media = mean(Nota, na.rm = TRUE),
    Desviacion_Tipica = sd(Nota, na.rm = TRUE),
    Percentil75 = quantile(Nota, 0.75, na.rm = TRUE),
    Maximo = max(Nota, na.rm = TRUE)
  ) %>%
  mutate(across(where(is.numeric), ~ round(.x, 2))) #Redondeamos a dos decimales.
tabla_resumen
```

```
## # A tibble: 5 x 8
##   Asignatura Minimo Percentil25 Mediana Media Desviacion_Tipica Percentil75
##   <chr>       <dbl>       <dbl>   <dbl> <dbl>             <dbl>       <dbl>
## 1 ALG           1.9        6       6.72  6.41              1.65        7.5
## 2 ANM           3.4        6.97    7.9   7.61              1.45        8.5
## 3 DCS           3          6.79    7.65  7.42              1.36        8.3
## 4 FP            3          3.53    6     5.77              2.25        7.25
## 5 MD            1.5        5.47    6.4   6.12              1.99        7.41
## # i 1 more variable: Maximo <dbl>
```

Gráfico con ggpairs

```r
notas_grafico <- datos %>%
  select(ALG, ANM, FP, DCS, MD)

# 2. Representamos las relaciones cruzadas con ggpairs
ggpairs(notas_grafico,
        title = "Relaciones cruzadas entre las calificaciones de las asignaturas",
        lower = list(continuous = wrap("points", alpha = 0.6, size = 1.5)),
        upper = list(continuous = wrap("cor", method = "pearson")))
```

```
## Warning: Removed 2 rows containing non-finite outside the scale range
## (`stat_density()`).

## Warning: Removed 2 rows containing missing values

## Warning: Removed 42 rows containing missing values

## Warning: Removed 2 rows containing missing values
## Removed 2 rows containing missing values

## Warning: Removed 2 rows containing missing values or values outside the scale range
## (`geom_point()`).

## Warning: Removed 2 rows containing non-finite outside the scale range
## (`stat_density()`).

## Warning: Removed 42 rows containing missing values

## Warning: Removed 2 rows containing missing values
## Removed 2 rows containing missing values

## Warning: Removed 42 rows containing missing values or values outside the scale range
## (`geom_point()`).
## Removed 42 rows containing missing values or values outside the scale range
## (`geom_point()`).

## Warning: Removed 42 rows containing non-finite outside the scale range
## (`stat_density()`).

## Warning: Removed 42 rows containing missing values
## Removed 42 rows containing missing values

## Warning: Removed 2 rows containing missing values or values outside the scale range
## (`geom_point()`).
## Removed 2 rows containing missing values or values outside the scale range
## (`geom_point()`).

## Warning: Removed 42 rows containing missing values or values outside the scale range
```

```
## (`geom_point()`).
```

```
## Warning: Removed 2 rows containing non-finite outside the scale range
## (`stat_density()`).
```

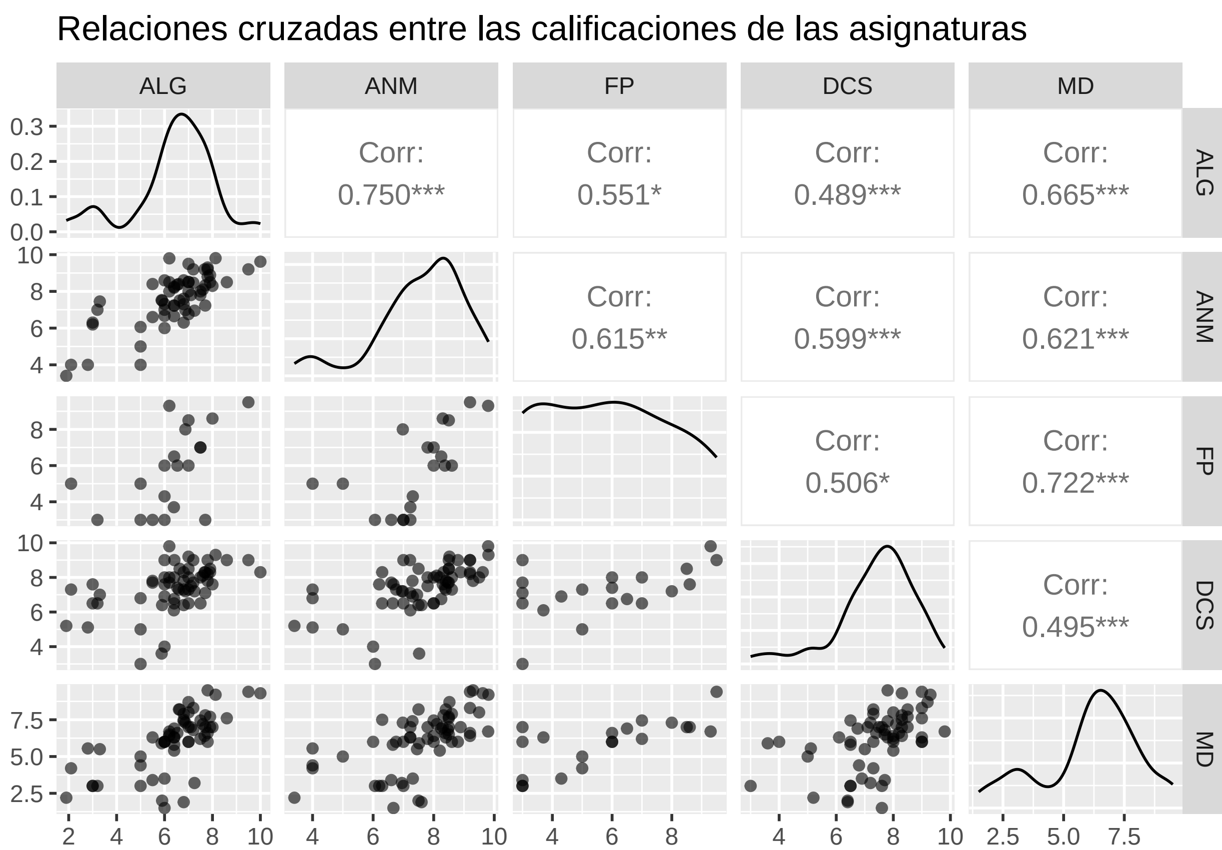
```
## Warning: Removed 2 rows containing missing values
```

```
## Warning: Removed 2 rows containing missing values or values outside the scale range
## (`geom_point()`).
## Removed 2 rows containing missing values or values outside the scale range
## (`geom_point()`).
```

```
## Warning: Removed 42 rows containing missing values or values outside the scale range
## (`geom_point()`).
```

```
## Warning: Removed 2 rows containing missing values or values outside the scale range
## (`geom_point()`).
```

```
## Warning: Removed 2 rows containing non-finite outside the scale range
## (`stat_density()`).
```



Determinar matrices de correlación

```r
#Matrices correlación
cat("MATRICES DE CORRELACIÓN \n")
```

```
## MATRICES DE CORRELACIÓN
```

```r
notas_grafico <-  notas_grafico %>% na.omit()
cor_pearson  <- round(cor(notas_grafico, method = "pearson"), 2)
cor_pearson
```

```
##      ALG  ANM   FP  DCS   MD
## ALG 1.00 0.71 0.55 0.33 0.80
## ANM 0.71 1.00 0.61 0.56 0.62
## FP  0.55 0.61 1.00 0.51 0.72
## DCS 0.33 0.56 0.51 1.00 0.50
## MD  0.80 0.62 0.72 0.50 1.00
```

```r
cor_spearman  <- round(cor(notas_grafico, method = "spearman"), 2)
cor_spearman
```

```
##      ALG  ANM   FP  DCS   MD
## ALG 1.00 0.60 0.62 0.28 0.86
## ANM 0.60 1.00 0.70 0.53 0.58
## FP  0.62 0.70 1.00 0.46 0.73
## DCS 0.28 0.53 0.46 1.00 0.33
## MD  0.86 0.58 0.73 0.33 1.00
```

```r
#Las de covarianza
cat("MATRICES DE COVARIANZA \n")
```

```
## MATRICES DE COVARIANZA
```

```r
cov_pearson  <- round(cov(notas_grafico, method = "pearson"), 2)
cov_pearson
```

```
##      ALG  ANM   FP  DCS   MD
## ALG 2.68 1.58 2.03 0.81 2.23
## ANM 1.58 1.86 1.89 1.13 1.45
## FP  2.03 1.89 5.08 1.70 2.77
## DCS 0.81 1.13 1.70 2.22 1.27
## MD  2.23 1.45 2.77 1.27 2.90
```

```r
cov_spearman  <- round(cov(notas_grafico, method = "spearman"), 2)
cov_spearman
```

```
##       ALG   ANM    FP   DCS    MD
## ALG 34.84 20.96 21.38  9.61 29.80
## ANM 20.96 34.92 24.22 18.62 20.07
## FP  21.38 24.22 34.32 15.83 25.16
## DCS  9.61 18.62 15.83 34.84 11.37
## MD  29.80 20.07 25.16 11.37 34.76
```

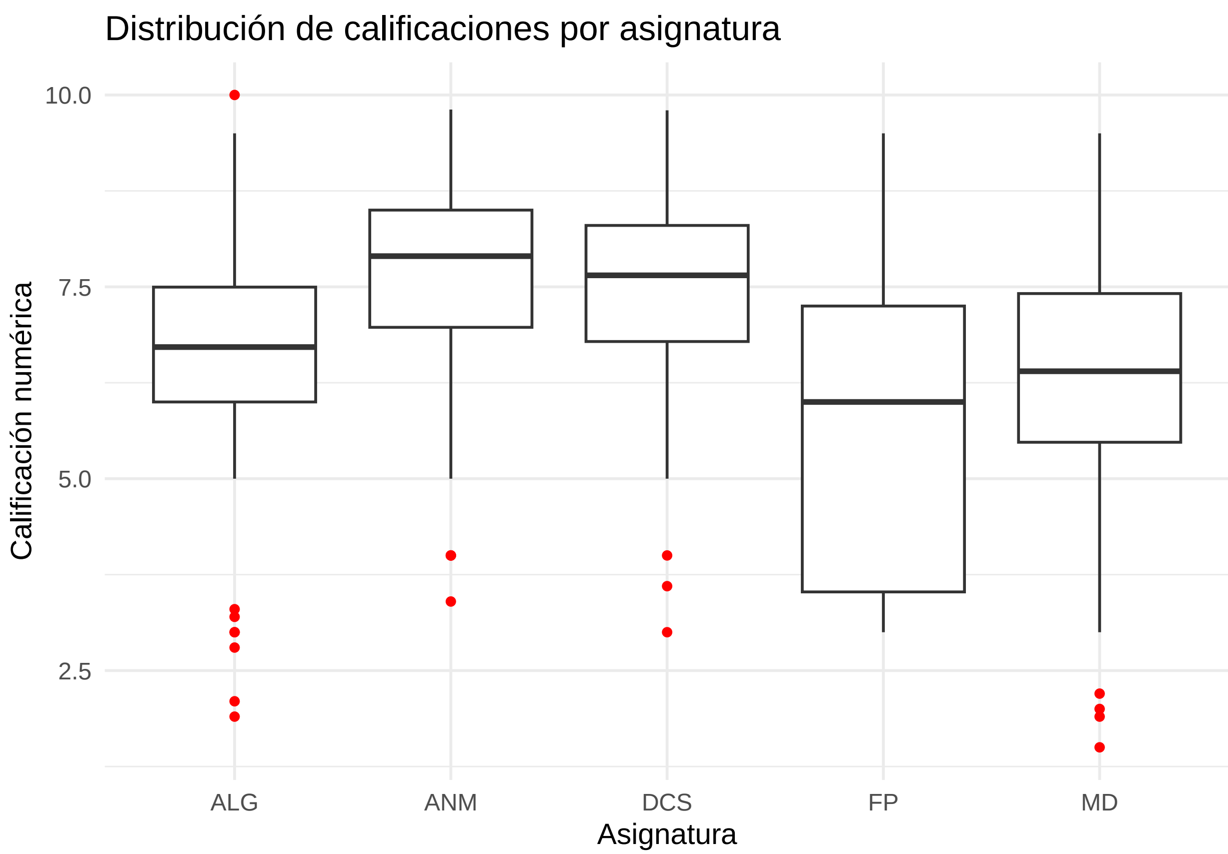
REPRESENTACIÓN BOXPLOTS

```r
datos_largos_notas <- datos %>%
  select(ALG, ANM, FP, DCS, MD) %>%
  pivot_longer(cols = everything(),
               names_to = "Asignatura",
               values_to = "Calificacion")

ggplot(datos_largos_notas, aes(x = Asignatura, y = Calificacion)) +
  geom_boxplot(outlier.color = "red", outlier.shape = 16) +
```

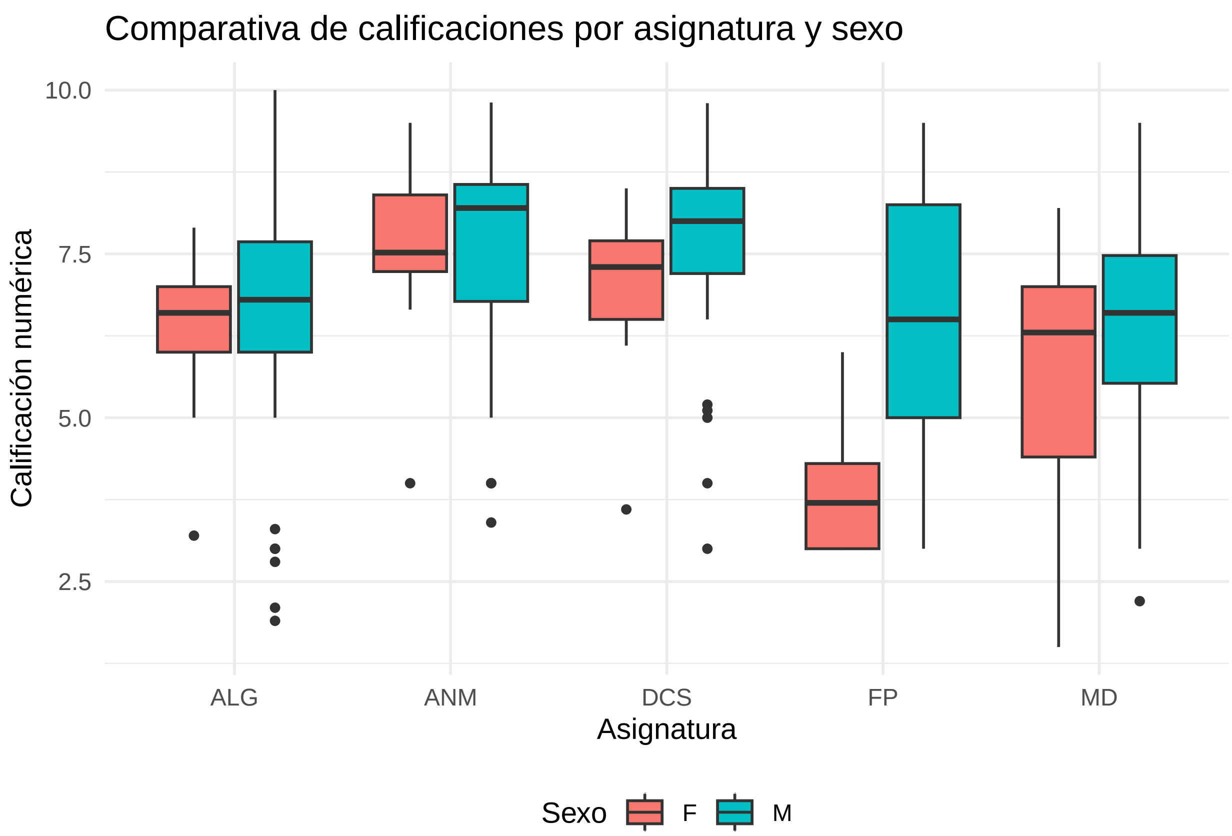
```
  labs(title = "Distribución de calificaciones por asignatura",
       x = "Asignatura",
       y = "Calificación numérica") +
  theme_minimal()
```

```
## Warning: Removed 50 rows containing non-finite outside the scale range
## (`stat_boxplot()`).
```




```
#PASO 10 (SEGÚN SEXO)
datos_largos_sexo <- datos %>%
  select(Sex, ALG, ANM, FP, DCS, MD) %>%
  pivot_longer(cols = c(ALG, ANM, FP, DCS, MD),
               names_to = "Asignatura",
               values_to = "Calificacion")
#Creación boxplot según sexo.
ggplot(datos_largos_sexo, aes(x = Asignatura, y = Calificacion, fill = Sex)) +
  geom_boxplot(outlier.shape = 16) +
  labs(title = "Comparativa de calificaciones por asignatura y sexo",
       x = "Asignatura",
       y = "Calificación numérica",
       fill = "Sexo") +
  theme_minimal() +
  theme(legend.position = "bottom")
```

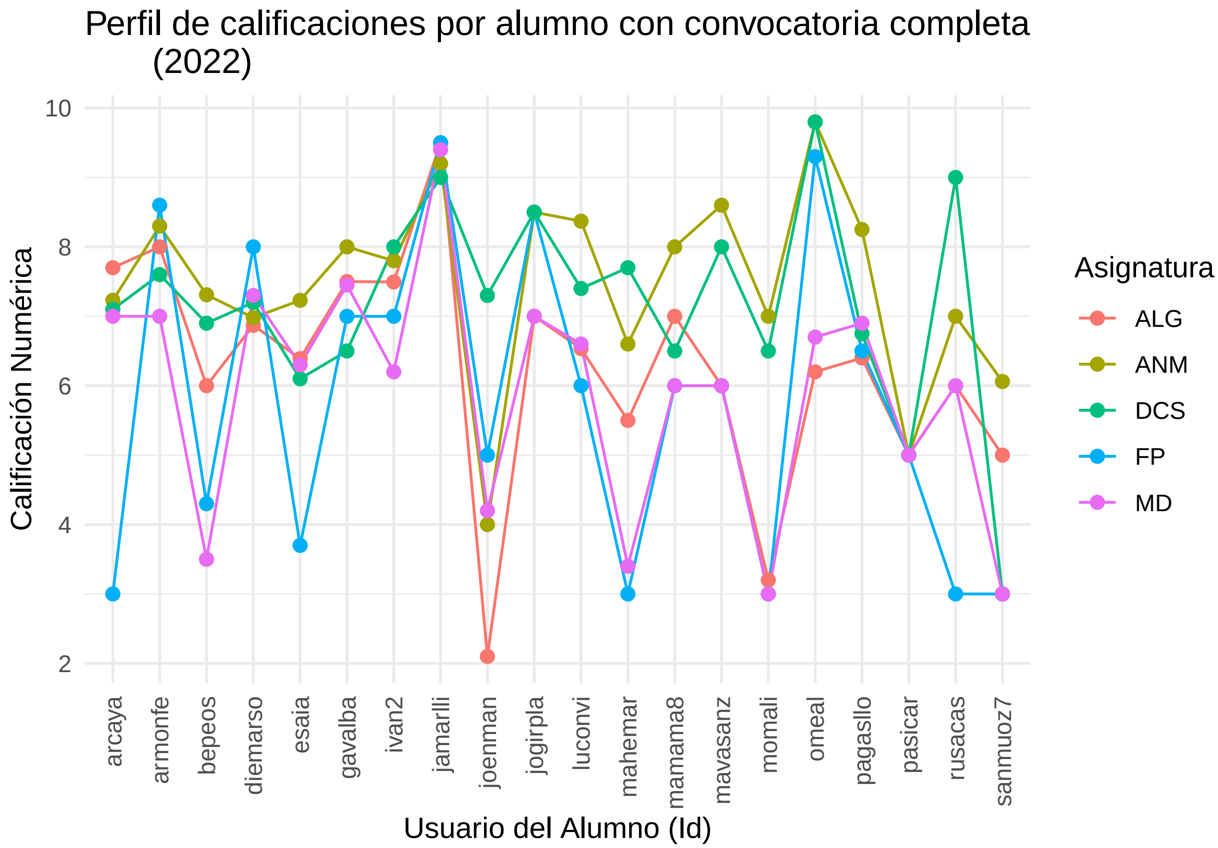
```
## Warning: Removed 50 rows containing non-finite outside the scale range
## (`stat_boxplot()`).
```



Selección alumnos y representación

```
#Seleccionamos alumnos con notas en todas las asignaturas
datos_completos <- datos %>%
  select(Id, ALG, ANM, FP, DCS, MD) %>%
  na.omit() %>%
  pivot_longer(cols = c(ALG, ANM, FP, DCS, MD),
               names_to = "Asignatura",
               values_to = "Calificacion")

ggplot(datos_completos, aes(x = Id, y = Calificacion, color = Asignatura,
                            group = Asignatura)) +
  geom_line() +
  geom_point(size = 2) +
  labs(title = "Perfil de calificaciones por alumno con convocatoria completa
       (2022)",
       x = "Usuario del Alumno (Id)",
       y = "Calificación Numérica",
       color = "Asignatura") +
  theme_minimal() +
  theme(axis.text.x = element_text(angle = 90, vjust = 0.5, hjust = 1))
```



Comparación con archivo de 2026

```
datos_2026 <- read_delim("data/NosAutoanalizamos2026 - NosAutoanalizamos.tsv",
                         delim = "\t",
                         escape_double = FALSE, #""
                         trim_ws = TRUE, #Fuera espacios en blanco
                         skip = 27) %>%
  filter(!is.na(Id))
```

```
## New names:
## Rows: 62 Columns: 23
## -- Column specification
## -------------------------------------------------------- Delimiter: "\t" chr
## (17): FechaNac(DD-MM-YYY), Id, Sex, Wr.Hnd, NW.Hnd, Fold, Clap, Smoke, A... dbl
## (4): Age, Pulse, Exer, Height num (1): Hwork lgl (1): ...22
## i Use `spec()` to retrieve the full column specification for this data. i
## Specify the column types or set `show_col_types = FALSE` to quiet this message.
## * `` -> `...22`
## * `` -> `...23`
```

```
datos_2026 <- datos_2026 %>%
  filter(Id != "martsobm")
#Mismas correcciones que en el de 2022(reemplazar NC por 3 y notas raras por NA)
datos_2026 <- datos_2026 %>%
  mutate(across(c(ALG, ANM, FP, DCS, MD), ~ {
    x <- as.character(.x)
    x <- str_replace(x, ",", ".")
    x[x == "NC"] <- "3"
```

```r
    as.numeric(x)
  }))
```

```
## Warning: There were 2 warnings in `mutate()`.
## The first warning was:
## i In argument: `across(...)`.
## Caused by warning:
## ! NAs introducidos por coerción
## i Run `dplyr::last_dplyr_warnings()` to see the 1 remaining warning.
```

```r
datos_2026$Sex[datos_2026$Sex == "Masculino"] <- "M"
datos_2026$Sex[datos_2026$Sex == "Hombre"]    <- "M"
datos_2026$Sex[datos_2026$Sex == "HOMBRE"]    <- "M"
datos_2026$Sex[datos_2026$Sex == "Femenino"]  <- "F"
datos_2026$Sex[datos_2026$Sex == "Mujer"]     <- "F"
datos_2026$Sex[datos_2026$Sex == "MUJER"]     <- "F"
```

RESPUESTAS A TODAS LAS PREGUNTAS

```r
# 1. Edad exacta con decimales del alumno más joven a fecha 01/02/2026
datos_2026 %>%
  mutate(Fecha_Nac = dmy(`FechaNac(DD-MM-YYY)`)) %>%
  filter(!is.na(Fecha_Nac)) %>%
  mutate(Edad_Exacta = as.numeric(difftime(ymd("2026-02-01"), Fecha_Nac, units = "days"))
         / 365.25) %>%
  summarise(Minima_Edad = min(Edad_Exacta, na.rm = TRUE)) %>%
  round(2)
```

```
## # A tibble: 1 x 1
##   Minima_Edad
##         <dbl>
## 1        18.1
```

```r
# 2. Alumnos que juntan trabajo y estudios
datos_2026 %>%
  mutate(Hwork_num = as.numeric(str_replace(as.character(Hwork), ",", "."))) %>%
  filter(Hwork_num > 0) %>%
  nrow()
```

```
## [1] 14
```

```r
# 3. Género más fumador
datos_2026 %>%
  filter(Smoke == "Si" | Smoke == "No") %>%
  filter(!is.na(Sex)) %>%
  count(Sex, Smoke)
```

```
## # A tibble: 3 x 3
##   Sex   Smoke     n
##   <chr> <chr> <int>
## 1 F     No       23
## 2 M     No       28
## 3 M     Si        7
```

```r
# 4. Valor medio y varianza Wr.Hnd
datos_2026 %>%
```

```r
  filter(!is.na(Sex)) %>%
  mutate(Wr.Hnd = as.numeric(str_replace(as.character(Wr.Hnd), ",", "."))) %>%
  group_by(Sex) %>%
  summarise(
    Media_Wr = mean(Wr.Hnd, na.rm = TRUE),
    Varianza_Wr = var(Wr.Hnd, na.rm = TRUE)
  )
```

```
## # A tibble: 2 x 3
##   Sex   Media_Wr Varianza_Wr
##   <chr>    <dbl>       <dbl>
## 1 F         18.0        3.72
## 2 M         20.0        1.73
```

```r
# 5. Buscar relación entre sexo y mano dominante
tabla_contingencia <- table(datos_2026$Sex, datos_2026$NW.Hnd)
chisq.test(tabla_contingencia) #Función chi-cuadrado de Pearson.
```

```
## Warning in chisq.test(tabla_contingencia): Chi-squared approximation may be
## incorrect
```

```
##
##  Pearson's Chi-squared test
##
## data:  tabla_contingencia
## X-squared = 2.7685, df = 2, p-value = 0.2505
```

El test de chi-cuadrado de Pearson devuelve un valor p = 0.2505. Al ser este valor superior al nivel estándar(0.05), se concluye que no hay relación entre el sexo del alumno y su mano dominante al escribir(independencia).

```r
# 6. Relación entre altura y Wr.Hnd
datos_2026 %>%
  mutate(
    Height = as.numeric(str_replace(as.character(Height), ",", ".")),
    Wr.Hnd = as.numeric(str_replace(as.character(Wr.Hnd), ",", "."))
  ) %>%
  select(Height, Wr.Hnd) %>%
  na.omit() %>%
  cor(method = "pearson")
```

```
##           Height    Wr.Hnd
## Height 1.0000000 0.4310869
## Wr.Hnd 0.4310869 1.0000000
```

Correlación moderada con tendencia positiva(0.43)

```r
# 7. Comprobar si Hampel detecta outliers en height
altura_vector <- datos_2026 %>%
  mutate(Height = as.numeric(str_replace(as.character(Height), ",", "."))) %>%
  filter(!is.na(Height)) %>%
  pull(Height)

hampel_analisis <- hampel(altura_vector, k = 3)
altura_vector[hampel_analisis$ind]
```

```
## [1] 163
```

El outlier detectado mediante el método Hampel es de una persona cuya altura son 163cm.

```r
# 8. Imputación generalizada en la asignatura de FP
mean(datos_2026$FP, na.rm = TRUE)
```

```
## [1] 4.4
```

```r
# 9. Definir variable PulseCat
datos_2026 %>%
  mutate(Pulse = as.numeric(as.character(Pulse))) %>%
  filter(!is.na(Pulse)) %>%
  mutate(PulseCat = if_else(Pulse >= 65, "PulseH", "PulseL")) %>%
  group_by(PulseCat) %>%
  summarise(Media_FP = mean(FP, na.rm = TRUE))
```

```
## # A tibble: 2 x 2
##   PulseCat Media_FP
##   <chr>       <dbl>
## 1 PulseH       5.1
## 2 PulseL       4.14
```

```r
# 10. Comprobar la siguiente frase: "El ritmo cardíaco está ligado con la
# cantidad de sangre que llega al cerebro, por tanto existirá una relación
# lineal directa esta esta variable y las calificaciones obtenidas en ALG"
# es cierta o no.
datos_2026 %>%
  mutate(Pulse = as.numeric(as.character(Pulse))) %>%
  select(Pulse, ALG) %>%
  na.omit() %>%
  cor(method = "pearson")
```

```
##           Pulse       ALG
## Pulse 1.0000000 0.1419435
## ALG   0.1419435 1.0000000
```

Hay poca correlación positiva entre el ritmo cardíaco y álgebra(0.14 aprox)